

Open Letter to the Clay Mathematics Institute

On the Formulation and Solution of the Navier–Stokes Problem

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To the Board of the Clay Mathematics Institute,

I write about the Navier–Stokes Millennium Problem: to propose a reformulation of it, to present a solution of the problem so reformulated, and to explain why I believe the reformulation is necessary rather than evasive. None of this is new. It was published in 2007, in a chapter of a book written with Johan Hoffman, and it has been put to the Institute before. What is new is the occasion. On 8 September 2026 OpenAI announced a solution of the problem as officially posed [27], and that announcement makes it worth setting out, once more and in one place, what I take the meaningful question to be, what answering it consists of, and what the announced solution does and does not settle.

I should say at the outset where I am writing from, because the proposal did not arrive from outside the subject. It is the end of a long piece of work, and the reformulation was reached by computing, not by reflecting on the problem statement.

1 Where this comes from: a computational programme, 1970s–2026

The 1970s: finite elements for elliptic problems. My work began with the finite element method for elliptic partial differential equations — error analysis, a priori and a posteriori estimates, and the question that has organised everything since: not whether a method converges in principle, but how large the error is in *this* computation, on *this* mesh, in a quantity somebody actually wants. For elliptic problems that question has a clean answer, and the answer is obtained by duality: the error in a functional of the solution is the residual of the computed solution weighted by the solution of an adjoint problem. Everything below is that idea, carried as far as it will go.

The 1980s: hyperbolic problems, and fluid flow. The methods were then extended to hyperbolic and convection-dominated problems, where the difficulties are of a different order: solutions develop sharp layers and discontinuities, Galerkin’s method loses stability, and stabilisation has to be supplied without destroying the error control. Streamline diffusion and least-squares stabilisation of the residual came out of that period, and with them the first com-

putations of fluid flow in which the dissipation was generated by the scheme in proportion to the residual rather than prescribed by the person running it.

The 1990s and 2000s: turbulent flow, adaptivity, error control. Carried to the Navier–Stokes equations at high Reynolds number, that programme produced the result which forced the reformulation. Computed turbulent solutions do not converge pointwise under mesh refinement — two meshes give two velocity fields that differ everywhere and go on differing — while *mean-value outputs* such as drag and lift converge to a few per cent, with the error bounded by duality against a computed adjoint solution rather than estimated by comparing successive meshes. That is not a defect of the computations. It is what the equations do, and once one has seen it repeatedly it is no longer possible to regard pointwise regularity as the property a formulation should ask about.

2007: the book, and the chapter. This was set out in *Computational Turbulent Incompressible Flow* [35], written with Johan Hoffman and published by Springer in 2007 as volume 4 of *Applied Mathematics: Body and Soul*. One chapter — Chapter 14 — is given to the Clay problem: it argues that the official formulation asks about the wrong property, proposes the reformulation in terms of output wellposedness, and presents computed solutions as its answer. What follows in this letter is that chapter’s content, recalled and brought up to date. The earlier statement of the approach is in [26], the residual-based dissipation in [18], the parameter-free adaptive formulation in [19], the resolution of d’Alembert’s paradox in [36], and a theory of flight built on the same computations in [37].

2007–2026: compressible flow, and where the programme now stands. The development since has gone to compressible flow and to aeronautical Reynolds numbers, and it has been carried substantially by Johan Jansson. The adaptive methodology was built into software — Unicorn, for adaptive simulation of turbulent flow and fluid–structure interaction [20], within a framework for massively parallel adaptive finite element computational fluid dynamics [21], and then FEniCS-HPC, for automated predictive high-performance finite element computing with aerodynamics applications [22] and coupled multiphysics [23], with parallel-in-time computation of turbulent flow in [24] — and applied at aeronautical Reynolds numbers in the DLR-F11 aircraft case, time-resolved and adaptively refined [25], and in the parameter-free formulation of [19]. The methodology now takes a leading role in the fluid mechanics workshops where computational predictions are compared against measurement. The argument of this letter is therefore not a position held in isolation. It is the reading of the Navier–Stokes equations that a working computational programme has arrived at, and the programme has been tested in the ordinary way, against experiment, for two decades.

2 The reformulation

Problem. *Constructive existence and qualitative uniqueness of weak solutions to the incompressible Navier–Stokes equations with small viscosity, that is at high Reynolds number. Given data: a domain $\Omega \subset \mathbb{R}^3$, initial data u_0 on Ω , a forcing f on $\Omega \times (0, T)$, and boundary conditions on $\partial\Omega$.*

Both adjectives carry weight, and each is weaker than the classical demand in one respect and stronger in another. In both the trade is the same: less is asked of the pointwise field, more of what can be done with it.

Constructive existence, as against existence only. Classical existence asserts that the solution set is non-empty. Leray’s theorem [32, 39] is of that kind: obtained by compactness, it establishes that a weak solution exists and says nothing about which one, what it looks like, or how to reach it. For *small* data — the low Reynolds number case for suitably scaled problems — global existence of a *smooth* solution has been known for sixty years [1, 2], so the difficulty lies entirely at the other end, which is the end the reformulation addresses. For a problem whose whole difficulty is that the solutions are complicated, that is close to no information. I ask instead that existence be established *by exhibition*, and concretely that five things be produced rather than asserted: the mesh, with its cell count and smallest cell; the discretisation carrying the solution, here continuous piecewise linear in space and time with least-squares control of the residual; the residual itself, measured in the norm the error bound uses; the computed stability factor for each output; and then, for each output — the drag of a body, the lift of a wing, a dissipation rate, a pressure distribution — a value with a bound on its error. That is what the solution *de facto* is. Ask a non-constructive existence theorem for the drag and there is nothing there to ask.

There is more to this than error control. A computed solution is an elaborate explicit construction of a velocity–pressure field, some millions of numbers on a spacetime mesh, every one available — and such an object can be taken apart. One can ask it where vorticity concentrates and in what shape, how the velocity gradient decomposes at a point, where energy passes from large scales to small and through what structures, where the dissipation actually sits as against where a model would put it. None of these questions can be put to a solution known only to exist. A solution concept whose objects can be inspected generates conjectures; one whose objects cannot be exhibited does not. The triple decomposition of the velocity gradient [16], and the energy stability analysis built on it [17], came out of looking closely at resolved velocity fields, and at measured ones [8].

Qualitative uniqueness, as against uniqueness only. Classical uniqueness asks that the data determine the field pointwise. At high Reynolds number this is false, and not marginally. What the data *do* determine is the qualitative content, and concretely this is what two computations on independently generated meshes are found to share: the drag agrees to a few per cent [38] and the lift likewise; the separation line sits in the same place; the wake is organised the same way and sheds at the same frequency; the total dissipation rate is the same. What they do not share is the velocity, which at any chosen point differs by order one and goes on differing under refinement.

Qualitative uniqueness is the demand that this be so, and that the line between the two lists be drawn *quantitatively* rather than by assertion. That last clause is what keeps it from being a soft version of uniqueness. The adjoint solution attaches to each functional a computed stability factor: where it is moderate the output is determined by the data to a stated precision, and refinement where the adjoint is large will improve it; where the factor is large the output is not determined, and no refinement will make it so. The claim is falsifiable one functional at a time, and a functional can fail it.

Three further things distinguish this from the official statement. *The domain is given*, not $\mathbb{R}^3$ with decay — real flows occupy bounded domains, and the boundary is where drag and lift are produced. *Small viscosity is part of the problem*, not a parameter fixed in advance: $\nu \rightarrow 0$ is the limit of interest and T is finite, which is the difference between asking about one Reynolds number at a time and asking about turbulence. And *weak solutions are asked for, smoothness is not* — by the Onsager theorem [30, 34] non-smoothness is not a defect to be removed but the mechanism by which a turbulent flow dissipates energy at a rate independent of viscosity — the dissipation anomaly, or zeroth law, observed since Taylor [10] and still under active examination [9]. Requiring smoothness excludes the solutions that occur.

3 The solution

Solution. *Computed solutions, as a form of constructed solution, for a representative choice of specific data — constructive existence by exhibition, qualitative uniqueness by the adjoint.* For given Ω , u_0 , f and boundary conditions, exhibit a computed weak solution together with quantitative control of the outputs: a residual small in the relevant norm, an error bound for each output obtained by duality from a computed solution of the adjoint equations, and a demonstration that the output converges under refinement while the velocity field does not.

Wellposedness is a property of outputs, not of solutions. This is the point on which the reformulation turns. Hadamard’s criterion has three parts — existence, uniqueness, continuous dependence on the data — and the third carries the physical content, because data are never exact. For a turbulent solution the pointwise velocity is not determined: perturb the data in the last decimal and the field is different everywhere within a short time. In that sense the solution is not wellposed and never will be. But nobody measures the pointwise velocity of a turbulent flow, and no engineering question depends on it. What is measured are mean-value outputs,

$$M(u) = \int_Q u \cdot \psi \, dx \, dt, \quad (1)$$

and the question of wellposedness must be asked of *them*. The claim is that $M(u)$ is stable for turbulent solutions even though u is not.

The dual solution is a weight function connecting residual to output. Output quality is not assessed by inspection; it is computed. The error in an output is, to leading order, the residual of the computed solution tested against a dual weight function φ ,

$$M(\hat{u}) - M(\hat{U}) \approx \int_0^T \int_\Omega R(\hat{U}) \cdot \varphi \, dx \, dt, \quad (2)$$

so φ is what converts a residual — which one has, and can measure — into an error in the quantity one cares about. Where φ is large the residual matters and the mesh must be refined; where it is small the residual is harmless however large; and its size overall is the stability factor that says whether the output is determined at all. This yields an a posteriori bound

$$|M(u) - M(U)| \leq \int_Q R(U, P) \cdot \varphi \, dx \, dt \leq S \|R(U, P)\|_{-1}, \quad (3)$$

in which S is not assumed but computed.

And φ is obtained by solving a definite problem: the Navier–Stokes equations linearised about the computed solution, run *backward in time* from the output functional as data at $t = T$. This is the same object, and the same computation, as the backward pass in the training of a neural network — the adjoint of a forward evolution, propagating sensitivity from an output to everything that fed it. Fluid mechanics and machine learning are doing the same linear algebra in opposite directions through the same trajectory.

Why the backward problem is stable for a turbulent solution. That the dual problem can be solved at all is the crux, and it is where intuition points the wrong way. A linearisation about a chaotic trajectory has exponentially growing modes; run backward, one expects φ to blow up and the whole assessment to be vacuous. It does not, and the reason is the oscillation. A turbulent solution does not grow exponentially; it *oscillates*, and it oscillates precisely *because* it is the result of exponential growth followed by exponential decay — a structure is stretched and amplified, becomes unstable, breaks down and dissipates, and the next takes its place. The linearised coefficients along such a trajectory change sign and structure repeatedly, so the growth φ accumulates over one interval is largely undone over the next: what would be a monotone exponential over a smooth field becomes a product of factors alternately greater and less than one. Over a *smooth* field there is nothing to cancel, and it grows.

So turbulence stabilises the outputs that smoothness destabilises, and by the same mechanism that makes it unpredictable pointwise. Smooth solutions are not wellposed. Turbulent solutions are wellposed in mean-value outputs. The physically meaningful solution concept is the one the official formulation treats as the failure case.

Turbulence is a necessity, not a possibility. It is worth adding why the third case is not one option among several. High Reynolds number flow starting out laminar develops shear instabilities with sharp velocity gradients, which are eventually damped by viscosity in turbulent flow. All high Reynolds number laminar flow is thus unstable and cannot continue to exist over time; it turns turbulent in order to continue to exist. Turbulence is what the equations must do.

4 Cases of computed solutions, with assessment of uniqueness

The method is General Galerkin, G2: a stabilised finite element method of Galerkin least-squares type [15], continuous piecewise linear in space and time, cG(1)cG(1), with least-squares control of the residual. The stabilisation is not a subgrid model — no eddy viscosity is prescribed, no wall function, no closure, no constant fitted to measurement. The dissipation is whatever the residual makes it, which means it is generated by the flow being computed rather than supplied by the person computing it. That matters for the present argument specifically: a computation carrying a fitted turbulence model could not be used to say what Navier–Stokes contains, because the turbulence would have been put in by hand.

Across the cases [35, 18, 19, 26, 25] the pattern is the one the reformulation predicts. The velocity field fails to converge under refinement and is not expected to. The mean-value output converges to a few per cent, and the error in it is bounded by (3), not estimated after the fact.

Mean drag and mean lift for turbulent flow past a bluff body come out to a few per cent on meshes of a few hundred thousand points — orders of magnitude coarser than a direct simulation resolving the Kolmogorov scale would require. And the adjoint supplies the assessment of uniqueness directly: where its stability factors are moderate the output is unique in the sense that matters; where they are large the output is not predictable and refinement will not make it so. That last case is not a failure of the method but an answer — the equations do not determine that quantity, and no computation should be believed which claims otherwise. Further material is collected under the label *clay problem* [40].

5 The official formulation recalled

The problem as set out by Fefferman [31] concerns

$$\dot{u} + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad (4)$$

on $\mathbb{R}^3$ or a periodic box, with $u(\cdot, 0) = u^0$ smooth and divergence free, and asks whether solutions remain smooth for all time or break down. I recall it here rather than at the start, because a formulation is best judged against one that has a solution.

The third case has no name in the formulation. Turbulent flow is not smooth in any useful sense: the gradients scale as

$$|\nabla u| \sim \nu^{-1/2}, \quad (5)$$

so at $\nu = 10^{-6}$ they are of order 10^3 times the velocity scale. Nor is it singular: the velocity stays bounded. The two halves are one scaling seen from either end,

$$\eta \sim \left(\frac{\nu^3}{\varepsilon}\right)^{1/4} \sim \nu^{3/4}, \quad \delta u(\eta) \sim (\varepsilon \eta)^{1/3} \sim \nu^{1/4}, \quad \frac{\delta u(\eta)}{\eta} \sim \nu^{-1/2}, \quad (6)$$

the first two being the Kolmogorov–Obukhov scaling [3, 4] — refined by the intermittency corrections of [5, 6] and the multifractal picture of [7] — and the middle relation being Hölder continuity with exponent $1/3$, the Onsager exponent. As $\nu \rightarrow 0$ the structures refine without limit while the velocity stays bounded at every scale: the gradient diverges, so the solution is not smooth, and the velocity does not, so it is not singular. A Hölder- $1/3$ field is exactly the object that is neither differentiable nor unbounded, which is why no amount of regularity theory reaches it. By the Onsager theorem [30, 34], proved after the partial results of [11, 33, 12, 13], this category has occupants, and a local energy equation can be written in which the non-smoothness is the direct cause of the dissipation [14].

The Institute’s own description of the problem opens on turbulence: waves follow the boat across the lake, turbulent air currents follow the aircraft, and the challenge is to unlock the secrets hidden in the equations. That is accurate about where the secrets are. The formal statement that follows contains no turbulence; it asks whether the velocity stays finite. Between the description and the problem the subject has been sorted out — not by oversight in the description, which names the phenomenon correctly, but by the formalisation, which admits no predicate that could distinguish a turbulent solution from a laminar one.

The alternatives are not complementary. Fefferman’s four statements do not form two pairs of negations. (A) and (B) are posed with $f \equiv 0$ and quantify over *all* admissible data: does every datum give a global smooth solution? (C) and (D) ask instead for the *existence* of admissible data *together with an admissible smooth forcing* f for which smoothness fails. So (C) is not $\neg(\text{A})$. A blowup driven by an external force leaves (A) untouched: it does not show that a fluid tears itself into a singularity, only that something can be done to it that does.

6 The announced solution, and the viscosity test

On 8 September 2026 OpenAI announced a proof that a singular solution exists [27], obtained with ten thousand agents and of the order of 3×10^{11} output tokens [28]. I take it to be a serious piece of work and I do not dispute it. If it stands, the Clay problem as posed is solved. Two observations follow, and the second is the substance of this letter.

What it is a solution of. Theorem 1.1 asserts that for every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty))$, a compact set K , and smooth fields u, p on $\mathbb{R}^3 \times [0, 1]$ solving (4) with $u(\cdot, 0) = 0$, supported in K , with $\sup_t \|u(t)\|_{L^2} < \infty$ and $\limsup_{t \uparrow 1} \|u(t)\|_{L^\infty} = \infty$. The flow starts from rest, so everything in it comes from the force — and the order of construction is the point. The velocity and pressure are built first, to have the required growth, and the force is defined as what is left over, by making its momentum residual the restriction of a force in C_c^∞ . The singularity is therefore supplied to the equations rather than generated by them, and the admissible class is broad enough to let it be supplied. That the class carries the result is visible from the other side too: replacing the C^∞ force by a spatially real-analytic one rules the mechanism out [29]. The paper states its own scope in just these terms — it establishes alternative (C), with the periodic case giving (D). It does not claim (A) or (B), and by the previous section it could not.

A shift in what a proof is, and why it bears on the reformulation. Something larger than the Navier–Stokes problem is visible here. With AI, mathematics is undergoing a monumental shift from symbolic analysis towards constructive computing — from proofs establishing that an object exists to proofs that exhibit one. The announced solution is an example of it. It does not argue abstractly that a singular solution must exist: it *builds* one, a velocity and pressure field specified in advance to have the required growth, with the force then defined as the residual of that construction. Ten thousand agents and of the order of 3×10^{11} tokens produced an explicit object, not an existence argument.

That is the direction in which this letter asks the formulation to move, and it is why the reformulation is not an eccentric demand. Constructive existence is becoming the normal form of a mathematical result rather than a weaker substitute for the real thing. The difference between the announced construction and the one I offer is therefore not that one is constructive and the other abstract — both are constructions, and both are specific. It is what they construct: a force built to order to produce a chosen singularity, against a flow computed forwards from data a fluid actually has, with a bound on what may be asked of it. Both exhibit. Only one exhibits a fluid.

The viscosity test. The test is general: it applies to any construction that exhibits a singularity at prescribed forcing. Navier–Stokes rescales — if v solves the equations at viscosity one with forcing g , then

$$u(x, t) = a v(bx, ct), \quad a = \nu b, \quad c = ab, \quad f = a^2 b g, \quad (7)$$

solves them at viscosity ν — and the map is onto: *one* unit-viscosity blowup generates the entire family. A theorem asserting blowup “for every $\nu > 0$ ” is therefore not a family of results. It is one result, rescaled, containing exactly what its $\nu = 1$ member contains.

Everything then turns on what (7) does to the constants, and since it is the forcing class that the official formulation fixes, two normalisations bear on it. Hold the solution at size one, $a = 1$: then $b = c = \nu^{-1}$ and the forcing amplitude is $a^2 b = \nu^{-1}$, so the force required *blows up* as the viscosity falls and leaves any fixed admissible class — including the one Fefferman specifies, which fixes its constants once and for all. Or hold the forcing in a fixed class, $a^2 b = 1$: then $b = \nu^{-2/3}$ and

$$\text{amplitude } a = \nu^{1/3} \rightarrow 0, \quad \text{length } b^{-1} = \nu^{2/3} \rightarrow 0, \quad \text{time } c^{-1} = \nu^{1/3} \rightarrow 0. \quad (8)$$

The singularity shrinks to nothing: at $\nu = 10^{-6}$ the velocity reaches 10^{-2} on a scale of 10^{-4} before blowing up. Neither branch survives the limit. One exits the admissible data, the other converges to the zero solution, and there is no third option.

Set this beside the flow the problem is advertised as being about. Turbulence at small viscosity has bounded velocity at every scale and gradients growing like $\nu^{-1/2}$, realised by (6) on structures of size $\nu^{3/4}$ across which the velocity varies by $\nu^{1/4}$ — amplitude fixed, structure refining. The constructed family has amplitude vanishing like $\nu^{1/3}$, which is the opposite behaviour, or else forcing diverging like ν^{-1} , which is not a fluid at all. A singularity that exists at every fixed ν and disappears as $\nu \rightarrow 0$ is not a feature of high-Reynolds-number flow; it is a feature of a formulation which admits forces no mechanism supplies.

7 In closing

The material above has been before the Institute, in one form or another, for some twenty years. It was published in 2007 in Chapter 14 of [35], it rests on a computational record documented in refereed journals from [26] onward, and the programme it belongs to has continued since into compressible flow and aeronautical Reynolds numbers. I have not changed the argument, and nothing in the intervening twenty years has required me to.

What has changed is the actuality, and it has changed in two ways. The first is the shift described above: as mathematics moves from symbolic analysis to constructive computing, a formulation asking for constructive existence and qualitative uniqueness is no longer asking for something unusual. The second is that a solution to the official formulation has now been announced, and it is of exactly the kind the formulation invites: a force constructed to produce a chosen singularity, admissible by the letter of the statement, and vanishing in the limit where fluid mechanics lives. The announcement therefore does not close the subject. It demonstrates, more clearly than argument could, that the official alternatives can be discharged without bearing on turbulence — and that a formulation whose horns can be satisfied without touching its subject is the thing that needs attention.

I would be glad to supply computational detail, or to discuss the formulation with anyone the Board cares to involve.

Yours faithfully,

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